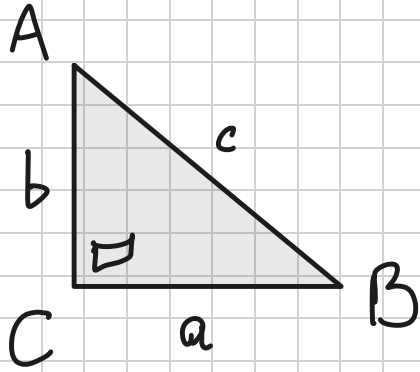


Week 1

P.7

Trigonometry review

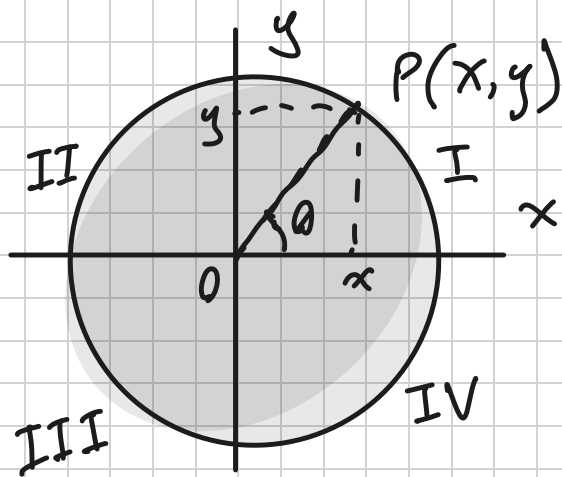


right triangle

$$\sin A = \frac{a}{c}, \quad \cos A = \frac{b}{c}$$

$$a^2 + b^2 = c^2 \quad \text{Pythagoras}$$

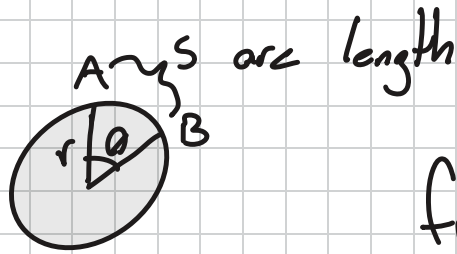
$$\frac{a^2}{c^2} + \frac{b^2}{c^2} = 1 \Rightarrow \sin^2 A + \cos^2 A = 1$$



Circle of radius 1

$$OP = 1$$

$$\sin \theta = y, \cos \theta = x$$



$$\theta_{\text{rad}} = \frac{s}{r}$$

full circle: 2π or 360°

$$\frac{s}{T} \bigg| \frac{A}{C}$$

transformation formula:

$$\theta_{\text{rad}} = \frac{\theta_{\text{degree}}}{360} \cdot 2\pi \Rightarrow \theta_{\text{rad}} = \pi \cdot \frac{\theta_{\text{deg}}}{180}$$

$$\theta_{\text{deg}} = \frac{\theta_{\text{rad}}}{\pi} \cdot 180$$

$$\sin 0 = 0, \quad \cos 0 = 1$$

$$\sin \frac{\pi}{2} = 1, \quad \cos \frac{\pi}{2} = 0$$

$$\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$\sin \frac{\pi}{6} = \frac{1}{2}, \quad \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}, \quad \cos \frac{\pi}{3} = \frac{1}{2}$$

$$\sin \frac{2\pi}{3} = -\frac{\sqrt{3}}{2}$$

$$\cos \frac{2\pi}{3} = -\frac{1}{2}$$

Complementary angles:

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos \theta$$

$\frac{\pi}{2} - \theta, \theta$ are called complementary

Supplementary angles:

$$\cos(\pi - \theta) = -\cos \theta$$

$$\sin(\pi - \theta) = \sin \theta$$

Addition / subtraction formulas

$$\sin(a \pm b) = \sin a \cos b \pm \sin b \cos a$$

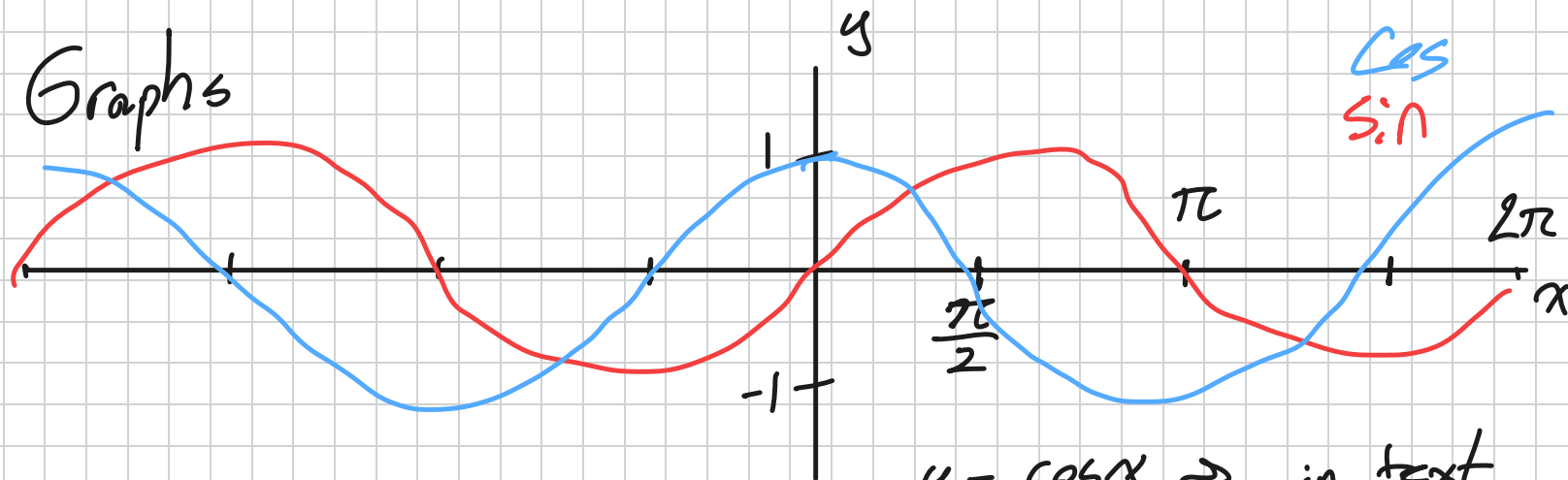
$$\cos(a \mp b) = \cos a \cos b \pm \sin a \sin b$$

Double angle formulas

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$$

Graphs



$\sin x, \cos x$ are defined over the real axis $\mathbb{R}$ $y = \cos x \rightarrow$ in text

Def ① $f: \mathbb{R} \rightarrow \mathbb{R}$ is even
if $f(-x) = f(x)$ for any $x \in \mathbb{R}$

② f is odd if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$

$$\begin{array}{ll} \sin x \text{ is odd} & \sin(-x) = -\sin x \\ \cos x \text{ is even} & \cos(-x) = \cos x \end{array}$$

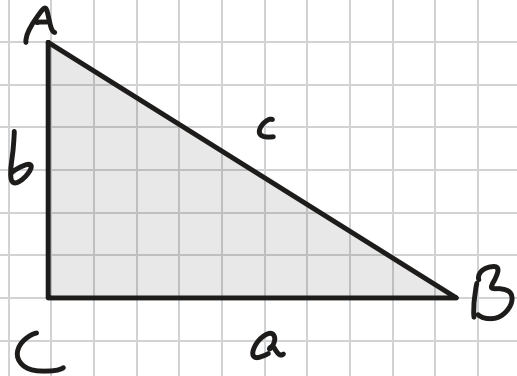
Example

Evaluate $\sin \frac{\pi}{12}$

note $\frac{\pi}{12} = \frac{\pi}{3} - \frac{\pi}{4}$

$$\sin \frac{\pi}{12} = \sin \left(\frac{\pi}{3} - \frac{\pi}{4} \right) = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{5} (\sqrt{3} - 1)$$

factor



$$\tan A = \frac{\sin A}{\cos A} = \frac{a}{b}$$

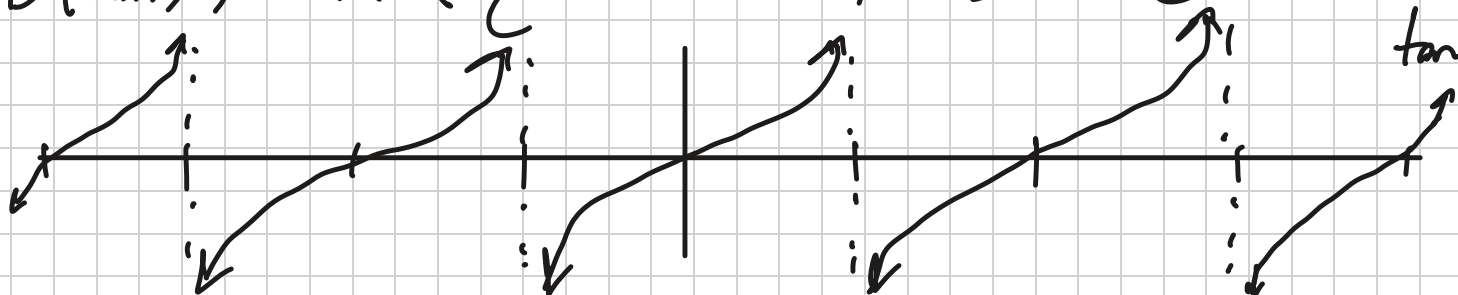
$$\tan x = \frac{\sin x}{\cos x}$$

$$\tan 0 = 0, \quad \tan \frac{\pi}{6} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}},$$

$$\tan \frac{\pi}{4} = 1, \quad \tan \frac{\pi}{3} = \sqrt{3}, \quad \tan \frac{\pi}{2} \text{ DNE}$$

exclude points

$$D(\tan x) = \mathbb{R} \setminus \left\{ \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \pm \frac{5\pi}{2}, \dots \right\}$$



all points
were lost
= 0

